1 A wooden cube of mass A is placed on an inclined plane. The cube is attached to a cylinder of mass B using string that passes over a pulley, as shown in Fig. 1.1.

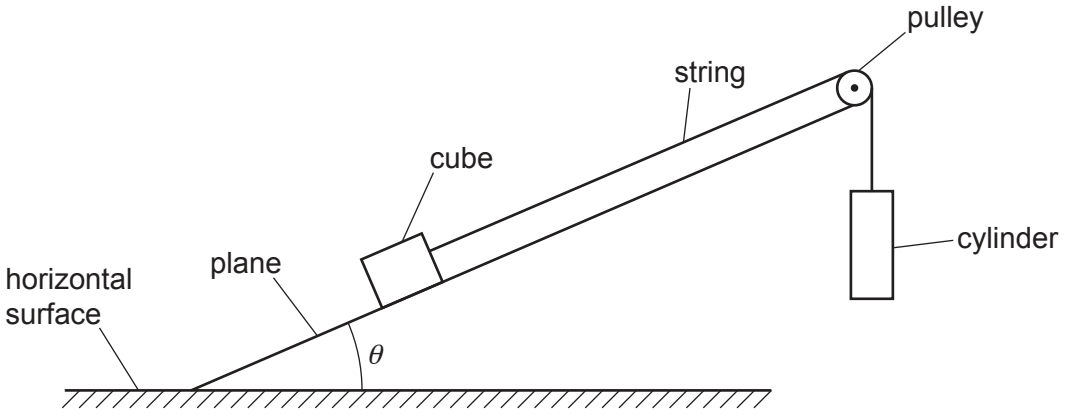

Fig. 1.1 (not to scale)

The angle between the plane and the horizontal surface is θ. Initially the cylinder is held at rest.

The cylinder is released. The time for the cylinder to fall a distance d is t.

It is suggested that t is related to θ by the relationship

$$\frac{2d}{t^2} = \frac{AH \sin\theta}{(A + B)} - \frac{KA}{(A + B)}$$

where H and K are constants.

Plan a laboratory experiment to test the relationship between t and θ.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for H and K.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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